Direct On Line [DOL] Forward Reverse Motor Very Basic MAPS Template (Electrical)

The Very Basic Direct On Line Forward Reverse (DOL_FR_VB) Motor MAPS template is designed to control a motor that is connected as a Forward Reverse DOL.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This MAPS template is a very basic forward reverse DOL.

This document describes the Very Basic Forward Reverse DOL Motor MAPS Template, which provides the lowest level of control over the Forward Reverse DOL motor. For details of its required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Very Basic Forward Reverse DOL Motor Functional Philosophy

The Forward Reverse DOL start forward output is enabled once all interlocks are ready and a start forward request is received the same for the start reverse request.

Field Interlocks

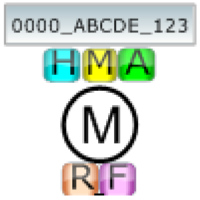
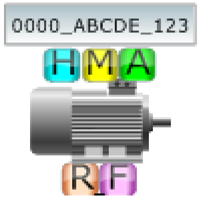
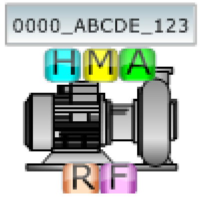
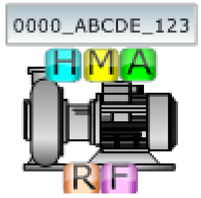
- Trip Feedback
- Fault Feedback

Interlock fault recovery process – once a field interlock is activated the faceplate will display a red indication, if the interlock is recovered and still latched in the PLC function block and the faceplate will show a green indication and the reset button on the faceplate will flash, pressing the Reset button will unlatch the interlock and display a grey indication and allow the DOL to become ready for operation.

Start and stop requests are initiated by selecting a mode on the DOL faceplate:

- **Auto mode** - the auto control program initiates a start and stop, normal PLC control.
- **Manual mode** - the operator can start and stop from the faceplate.
- **Hand mode** - the motor is in PLC bypass mode.

Typical Faceplate Graphics for the Very Basic Forward Reverse DOL Motor MAPS Template

FR DOL ISO	FR DOL Aerator	FR DOL Horizontal Mounted Motor Left	FR DOL PUMP LEFT	FR DOL PUMP RIGHT
				
H	Hand Mode			
M	Manual Mode			
A	Auto Mode			
F	Forward			
R	Reverse			
0000_ABCDE_123	Motor Control Name			

Graphics legend:

Line colour	Fill colour	DOL status
Black	Grey	Stopped
Grey	White	Stopping
Green	White	Starting
Black	Green	Running
Black	Yellow/Red	Fault Active

Signal Description		Agent Type	Very Basic
SCADA Control Words:			
SSCL 0	Control Word	**Marshal**	**x**

Signal Description		Agent Type	Very Basic
SCADA Status Words:			
SSSL 0	Status Word	**Marshal**	**x**
SSSL 1	Total Running Hours	**Analog**	**x**

Digital Inputs:		Very Basic
DI 0	Hand (On = Hand)	**x**
DI 1	Auto (On = Auto)	**x**
DI 2	Run Forward Feedback (On = Running Forward)	**x**
DI 3	Run Reverse Feedback (On = Running Reverse)	**x**
DI 4	Trip (On = Tripped)	**x**
DI 5	Fault (On = Fault)	**x**
Digital Outputs:		
DO 0	Device Start Forward	**x**
DO 1	Device Start Reverse	**x**

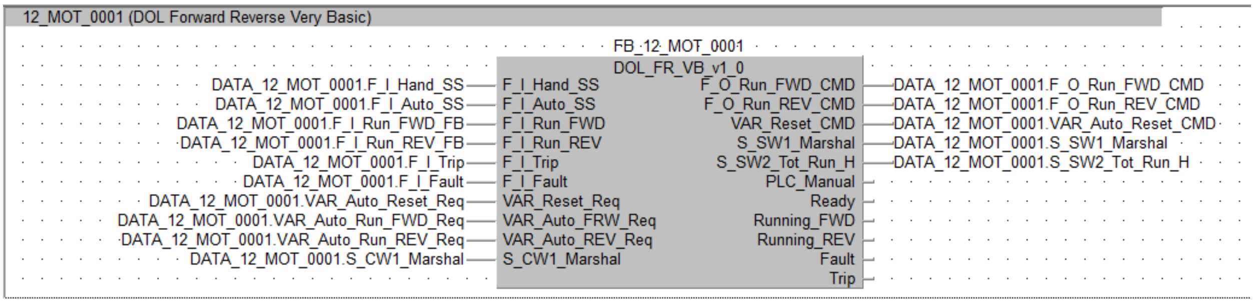
SCADA Control Word (Adroit Marshal Agent)	Event Logged	Very Basic

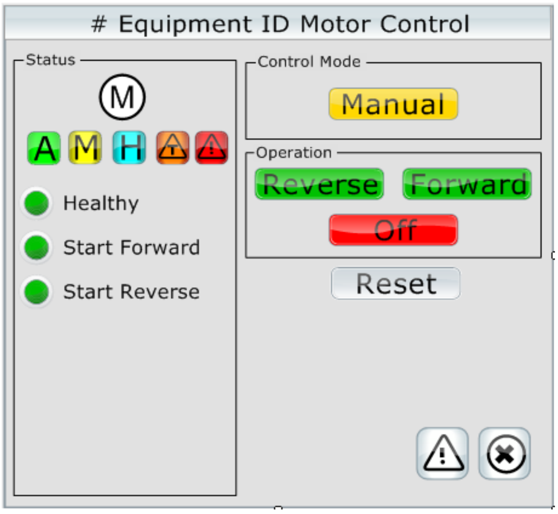
Bit 0	Alarms All Reset	x	x
Bit 1	Manual	x	x
Bit 2	Start Forward Command	x	x
Bit 3	Start Reverse Command	x	x
Bit 4	Stop Command	x	x
Bit 5	Spare		
Bit 6	Spare		
Bit 7	Spare		
Bit 8	Spare		
Bit 9	Spare		
Bit 10	Spare		
Bit 11	Spare		
Bit 12	Spare		
Bit 13	Spare		
Bit 14	Spare		
Bit 15	Spare		

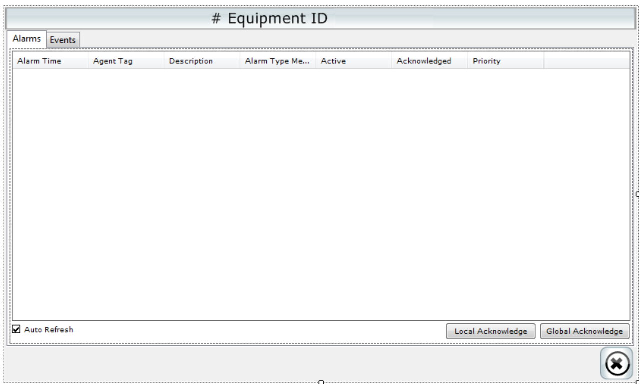
SCADA Status Word (Adroit Marshal Agent)		**Alarmed**	**Very Basic**
Bit 0	Hand		x
Bit 1	Auto		x
Bit 2	Manual		x
Bit 3	Ready		x
Bit 4	Starting		x
Bit 5	Running Forward		x
Bit 6	Running Reverse		x
Bit 7	Trip	x	x
Bit 8	Fault	x	x
Bit 9	Spare		
Bit 10	Spare		
Bit 11	Spare		
Bit 12	Spare		
Bit 13	Spare		
Bit 14	Spare		
Bit 15	Spare		

Resources	**Very Basic**
Digital Inputs	6
Digital Outputs	2
Analog Inputs	0
Analog Outputs	0
PLC Memory Steps	230
System Words Used	9
System Bits Used	76
SCADA SCAN Points	2

Template: Very Basic Forward Reverse DOL PLC Function Block



Template Graphic: Advanced Home Screen	Control	Descriptions
	Manual	Manual control by the operator
	Forward	Manually start Forward the DOL
	Reverse	Manually start Reverse the DOL
	Off	Manually stop the DOL
	Reset	Reset the DOL after a fault has been repaired
	Healthy	Healthy to start the DOL

Template Graphic: Advanced Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.